INFORMATION RETRIEVAL METRICS

$TP - True\ Positive$

$FP = False\ Positive$

$FN = False\ Negative$

$TN = True\ Negative$

ENROLMENT PROCEDURES

English Query 1: *"How to drop subjects?"*

$TP = 3$

$FP = 0$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+1}$$

$$Recall = 0.75$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.75}{1 + 0.75}$$

$$F1 - Score = 2 \times \frac{0.75}{1.75}$$

$$F1 - Score = 0.8571$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 556}{3 + 556 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 1: “Pano po magtanggal ng subject na kinuha ko?”

$TP = 2$

$FP = 1$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+1}$$

$$Recall = 0.6667$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.6667}{0.6667 + 0.6667}$$

$$F1 - Score = 2 \times \frac{0.4444}{1.3333}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 556}{2 + 556 + 1 + 1}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

Taglish QUERY 1: “Paano mag-drop ng subject?”

$TP = 2$

$FP = 1$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+1}$$

$$Recall = 0.6667$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.6667}{0.6667 + 0.6667}$$

$$F1 - Score = 2 \times \frac{0.4444}{1.3333}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 556}{2 + 556 + 1 + 1}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

English QUERY 2: How to shift/change curricular program?

$TP = 3$

$FP = 1$

$FN = 0$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+1}$$

$$Precision = 0.75$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.75 \times 1}{0.75 + 1}$$

$$F1 - Score = 2 \times \frac{0.75}{1.75}$$

$$F1 - Score = 0.8571$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 556}{3 + 556 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 2: “Pano po lumipat ng kurso?”

$TP = 3$

$FP = 0$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 557}{3 + 557 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

Taglish QUERY 2: “Paano mag-shift ng course?”

$TP = 3$

$FP = 0$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 557}{3 + 557 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

English QUERY 3: “How to enroll for the next semester?”

$TP = 3$

$FP = 0$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 557}{3 + 557 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

Tagalog QUERY 3: “Ano pong kailangan para makapag-enroll sa susunod na sem?”

$TP = 3$

$FP = 0$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 557}{3 + 557 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

Taglish QUERY 3: “Paano mag-enroll for next sem?”

$TP = 3$

$FP = 0$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 557}{3 + 557 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

English QUERY 4: “How to add or change subjects?”

$TP = 3$

$FP = 0$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+1}$$

$$Recall = 0.75$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.75}{1 + 0.75}$$

$$F1 - Score = 2 \times \frac{0.75}{1.75}$$

$$F1 - Score = 0.8571$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 556}{3 + 556 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 4: “Pano po magdagdag o magpalit ng subject?”

$TP = 2$

$FP = 1$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+1}$$

$$Recall = 0.6667$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.6667}{0.6667 + 0.6667}$$

$$F1 - Score = 2 \times \frac{0.4444}{1.3333}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 556}{2 + 556 + 1 + 1}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

Taglish QUERY 4: “Paano mag-add or mag-change ng subject after mag-enroll?”

$TP = 2$

$FP = 1$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 1}{0.6667 + 1}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Grades & Academic Standing

English QUERY 5: “How to remove a grade of 4?”

$TP = 3$

$FP = 0$

$FN = 2$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+2}$$

$$Recall = 0.6$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.6}{1 + 0.6}$$

$$F1 - Score = 2 \times \frac{0.6}{1.6}$$

$$F1 - Score = 0.75$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 555}{3 + 555 + 0 + 2}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

Tagalog QUERY 5: “May 4 po ako sa isang subject, pano po ito matatanggal?”

$TP = 2$

$FP = 1$

$FN = 2$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.5}{0.6667 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.3333}{1.1667}$$

$$F1 - Score = 0.5714$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 555}{2 + 555 + 1 + 2}$$

$$Accuracy = \frac{557}{560}$$

$$Accuracy = 0.9946$$

Taglish QUERY 5: “Paano mag-remove ng grade na 4?”

$TP = 2$

$FP = 1$

$FN = 2$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.5}{0.6667 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.3333}{1.1667}$$

$$F1 - Score = 0.5714$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 555}{2 + 555 + 1 + 2}$$

$$Accuracy = \frac{557}{560}$$

$$Accuracy = 0.9946$$

English QUERY 6: “How to complete an INC grade?”

$TP = 2$

$FP = 1$

$FN = 2$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.5}{0.6667 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.3333}{1.1667}$$

$$F1 - Score = 0.5714$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 555}{2 + 555 + 1 + 2}$$

$$Accuracy = \frac{557}{560}$$

$$Accuracy = 0.9946$$

Tagalog QUERY 6: “Pano ko po matatapos yung INC ko?”

$TP = 2$

$FP = 1$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 1}{0.6667 + 1}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Taglish QUERY 6: “May INC ako sa isang subject, paano ko siya ma-complete?”

$TP = 2$

$FP = 1$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+1}$$

$$Recall = 0.6667$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.6667}{0.6667 + 0.6667}$$

$$F1 - Score = 2 \times \frac{0.4444}{1.3333}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 556}{2 + 556 + 1 + 1}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

English QUERY 7: “How do I get off probation?”

$TP = 2$

$FP = 1$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 1}{0.6667 + 1}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 7: “Pano po makakaalis sa probation?”

$TP = 2$

$FP = 1$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 1}{0.6667 + 1}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Taglish QUERY 7: “Paano matanggal sa probation?”

$TP = 2$

$FP = 1$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 1}{0.6667 + 1}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Discipline & Student Conduct

English QUERY 8: “What do I do when I lose, forget my id or to retrieve confiscated?”

$TP = 1$

$FP = 1$

$FN = 0$

$TN = 558$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{1}{1+1}$$

$$Precision = 0.5$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{1}{1+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.5 \times 1}{0.5 + 1}$$

$$F1 - Score = 2 \times \frac{0.5}{1.5}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{1 + 558}{1 + 558 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 8: “Nawala ko po yung ID ko o kinuha ng guard, pano ko po makukuha ulit?”

$TP = 1$

$FP = 1$

$FN = 0$

$TN = 558$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{1}{1+1}$$

$$Precision = 0.5$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{1}{1+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.5 \times 1}{0.5 + 1}$$

$$F1 - Score = 2 \times \frac{0.5}{1.5}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{1 + 558}{1 + 558 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Taglish QUERY 8: “Na-confiscate yung ID ko, paano ko makukuha ulit? Pano rin pag naiwan ko?”

$TP = 1$

$FP = 1$

$FN = 0$

$TN = 558$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{1}{1+1}$$

$$Precision = 0.5$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{1}{1+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.5 \times 1}{0.5 + 1}$$

$$F1 - Score = 2 \times \frac{0.5}{1.5}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{1 + 558}{1 + 558 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

English QUERY 9: “What is the penalty for cheating in an exam?”

$TP = 2$

$FP = 0$

$FN = 2$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.5}{1 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.5}{1.5}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 556}{2 + 556 + 0 + 2}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

Tagalog QUERY 9: “Ano po ang parusa kapag nahuling nangongopya sa exam?”

$TP = 2$

$FP = 0$

$FN = 2$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.5}{1 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.5}{1.5}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 556}{2 + 556 + 0 + 2}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

Taglish QUERY 9: “Ano penalty pag nahuli kang nagcheat sa exam?”

$TP = 2$

$FP = 0$

$FN = 2$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.5}{1 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.5}{1.5}$$

$$F1 - Score = 0.6667$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 556}{2 + 556 + 0 + 2}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

English QUERY 10: “How to secure a good moral certificate?”

$TP = 3$

$FP = 0$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+1}$$

$$Recall = 0.75$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.75}{1 + 0.75}$$

$$F1 - Score = 2 \times \frac{0.75}{1.75}$$

$$F1 - Score = 0.8571$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 556}{3 + 556 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 10: “Pano po kumuha ng Good Moral?”

$TP = 3$

$FP = 0$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+1}$$

$$Recall = 0.75$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.75}{1 + 0.75}$$

$$F1 - Score = 2 \times \frac{0.75}{1.75}$$

$$F1 - Score = 0.8571$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 556}{3 + 556 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Taglish QUERY 10: “Paano mag-request ng Good Moral?”

$TP = 3$

$FP = 0$

$FN = 1$

$TN = 556$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+1}$$

$$Recall = 0.75$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.75}{1 + 0.75}$$

$$F1 - Score = 2 \times \frac{0.75}{1.75}$$

$$F1 - Score = 0.8571$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 556}{3 + 556 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Student Services & Records

English QUERY 11: “How to secure Transfer Credentials?”

$TP = 2$

$FP = 1$

$FN = 0$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 1}{0.6667 + 1}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 11: “Lilipat na po ako ng ibang unibersidad, pano po kukunin yung transfer credentials ko?”

$TP = 2$

$FP = 1$

$FN = 2$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.5}{0.6667 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.3333}{1.1667}$$

$$F1 - Score = 0.5714$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 555}{2 + 555 + 1 + 2}$$

$$Accuracy = \frac{557}{560}$$

$$Accuracy = 0.9946$$

Taglish QUERY 11: “Paano kumuha ng transfer credentials? Magta-transfer na kasi ako ng ibang university.”

$TP = 2$

$FP = 1$

$FN = 2$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+1}$$

$$Precision = 0.6667$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+2}$$

$$Recall = 0.5$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6667 \times 0.5}{0.6667 + 0.5}$$

$$F1 - Score = 2 \times \frac{0.3333}{1.1667}$$

$$F1 - Score = 0.5714$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 555}{2 + 555 + 1 + 2}$$

$$Accuracy = \frac{557}{560}$$

$$Accuracy = 0.9946$$

English QUERY 12: “How do I claim accident insurance?”

$TP = 2$

$FP = 0$

$FN = 0$

$TN = 558$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 558}{2 + 558 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

Tagalog QUERY 12: “Naaksidente po ako, pano po makakuha ng insurance?”

$TP = 2$

$FP = 0$

$FN = 0$

$TN = 558$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 558}{2 + 558 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

Taglish QUERY 12: “Paano mag-claim ng accident insurance?”

$TP = 2$

$FP = 0$

$FN = 0$

$TN = 558$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 1}{1 + 1}$$

$$F1 - Score = 2 \times \frac{1}{2}$$

$$F1 - Score = 1$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 558}{2 + 558 + 0 + 0}$$

$$Accuracy = \frac{560}{560}$$

$$Accuracy = 1$$

English QUERY 13: “How to obtain a vehicle or gate pass?”

$TP = 2$

$FP = 0$

$FN = 1$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+1}$$

$$Recall = 0.6667$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.6667}{1 + 0.6667}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Tagalog QUERY 13: “Pano po kumuha ng gate pass para makapasok yung sasakyan ko?”

$TP = 2$

$FP = 0$

$FN = 1$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+1}$$

$$Recall = 0.6667$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.6667}{1 + 0.6667}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Taglish QUERY 13: “Paano kumuha ng gate pass para makapasok yung vehicle ko sa campus?”

$TP = 2$

$FP = 0$

$FN = 1$

$TN = 557$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{2}{2+0}$$

$$Precision = 1$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{2}{2+1}$$

$$Recall = 0.6667$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{1 \times 0.6667}{1 + 0.6667}$$

$$F1 - Score = 2 \times \frac{0.6667}{1.6667}$$

$$F1 - Score = 0.8$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{2 + 557}{2 + 557 + 0 + 1}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

English QUERY 14: “How to avail the use of university rooms and facilities?”

$TP = 3$

$FP = 2$

$FN = 0$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+2}$$

$$Precision = 0.6$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6 \times 1}{0.6 + 1}$$

$$F1 - Score = 2 \times \frac{0.6}{1.6}$$

$$F1 - Score = 0.75$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 555}{3 + 555 + 2 + 0}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$

Tagalog QUERY 14: “Pano po manghiram ng room dito sa school para sa programa namin?”

$TP = 4$

$FP = 1$

$FN = 0$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{4}{4+1}$$

$$Precision = 0.8$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{4}{4+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.8 \times 1}{0.8 + 1}$$

$$F1 - Score = 2 \times \frac{0.8}{1.8}$$

$$F1 - Score = 0.8889$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{4 + 555}{4 + 555 + 1 + 0}$$

$$Accuracy = \frac{559}{560}$$

$$Accuracy = 0.9982$$

Taglish QUERY 14: “Paano mag-reserve ng room or facility sa PSU para sa event namin?”

$TP = 3$

$FP = 2$

$FN = 0$

$TN = 555$

$Precision$:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{3}{3+2}$$

$$Precision = 0.6$$

$Recall$:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{3}{3+0}$$

$$Recall = 1$$

$F1\ Score$:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 - Score = 2 \times \frac{0.6 \times 1}{0.6 + 1}$$

$$F1 - Score = 2 \times \frac{0.6}{1.6}$$

$$F1 - Score = 0.75$$

$Accuracy$:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{3 + 555}{3 + 555 + 2 + 0}$$

$$Accuracy = \frac{558}{560}$$

$$Accuracy = 0.9964$$